

17USATST6

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June 16, 2026

Problem

Prove that there are infinitely many triples (a, b, p) of positive integers with p prime, $a < p$, and $b < p$, such that $(a + b)^p - a^p - b^p$ is a multiple of p^3 .

Solution. There are infinitely many p such that $p \equiv 1 \pmod{3}$. Choose such a p .

Claim 1 — $(x^2 + x + 1)^2 \mid (x + 1)^p - 1^p - x^p$.

Proof. Let ω be a non real cube root of unity. We have, $(\omega + 1)^{6k+1} - 1 - \omega^{6k+1} = -\omega^2 - 1 - \omega = 0$. So, $x^2 + x + 1 \mid (x + 1)^p - 1^p - x^p$. Taking the derivative of the expression, we have, $p(x + 1)^{p-1} - px^{p-1}$. Again, $p(\omega + 1)^{6k} - p\omega^{6k} = p - p = 0$. So, $(x^2 + x + 1)^2 \mid (x + 1)^p - 1^p - x^p$. $\square$

Since we already know that $(a + b)^p \equiv a^p + b^p \pmod{p}$ due to Freshman's Dream, it is enough to show that there is a n such that $p \mid n^2 + n + 1$.

Let g be a primitive root $\pmod{p}$. Then, $n = g^{\frac{p-1}{3}}$ has order 3 $\pmod{p}$. So, $p \mid n^2 + n + 1$ and hence $p^3 \mid (n + 1)^p - n^p - 1^p$.